

What Is This Class???

HES 505 Fall 2025: Session 1

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Welcome to Space!!

Today's Plan

- Introductions
- What can we do with spatial data?
- Course logistics and resources
- Testing out RStudio and GitHub Classroom

Introductions

About you?

- Your preferred pronouns
- Where are you from?
- What do you like most about Boise?
- What do you miss most about “home”?
- What is your research?

About Me

- What I do
- My path to this point
- Why I teach this course



The background of the slide is a photograph of a coastal scene. In the foreground, several large, white icebergs float on a calm body of water. The water reflects the sky and the ice. In the background, there are low, dark mountains or hills under a vast, cloudy sky. The clouds are white and grey, with some blue visible between them. The overall color palette is cool, with blues, greys, and whites.

What can we do with spatial data?

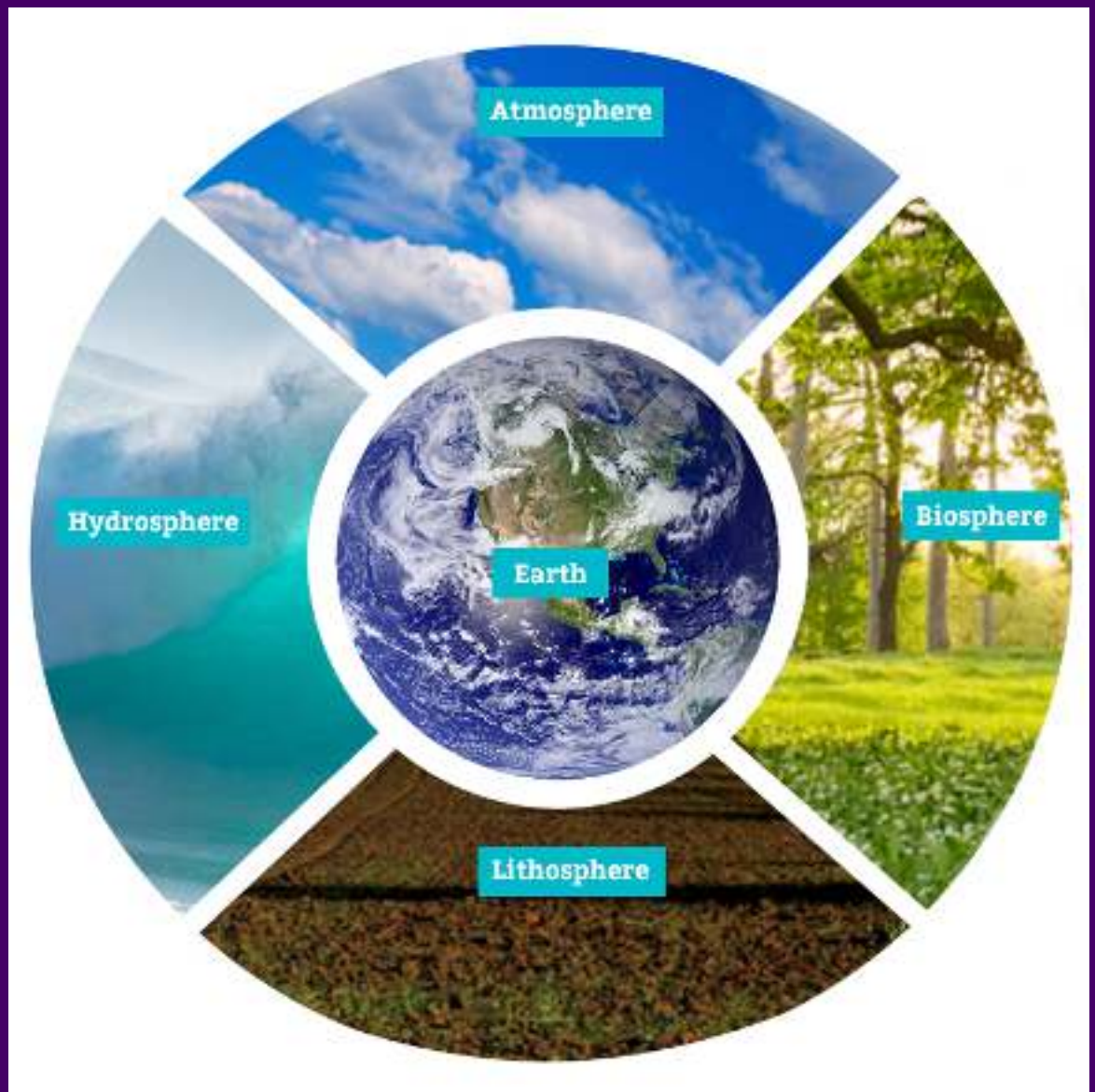
What is geography

- **Geo:** land, earth, terrain
- **Graph:** writing, discourse
- Tuan: **Space** (extent) and **Place** (location)
- Analysis of the effects of extent and location on events or features

Five Themes in Geography

- **Location**
 - Absolute location: latitude and longitude
 - Relative location: location in relation to other places
- **Place**
 - Physical place: natural features of a location
 - Human place: human-made features of a location
- **Region**
 - Formal region: region defined by specific characteristics
 - Functional region: region defined by a common function
 - Vernacular region: region defined by a common identity
- **Movement**
 - Migration: movement of people from one place to another
 - Trade: movement of goods and services between places
 - Communication: movement of information between places
- **Human-Environment Interaction**
 - Adaptation: how humans adapt to their environment
 - Modification: how humans change their environment
 - Dependence: how humans depend on their environment

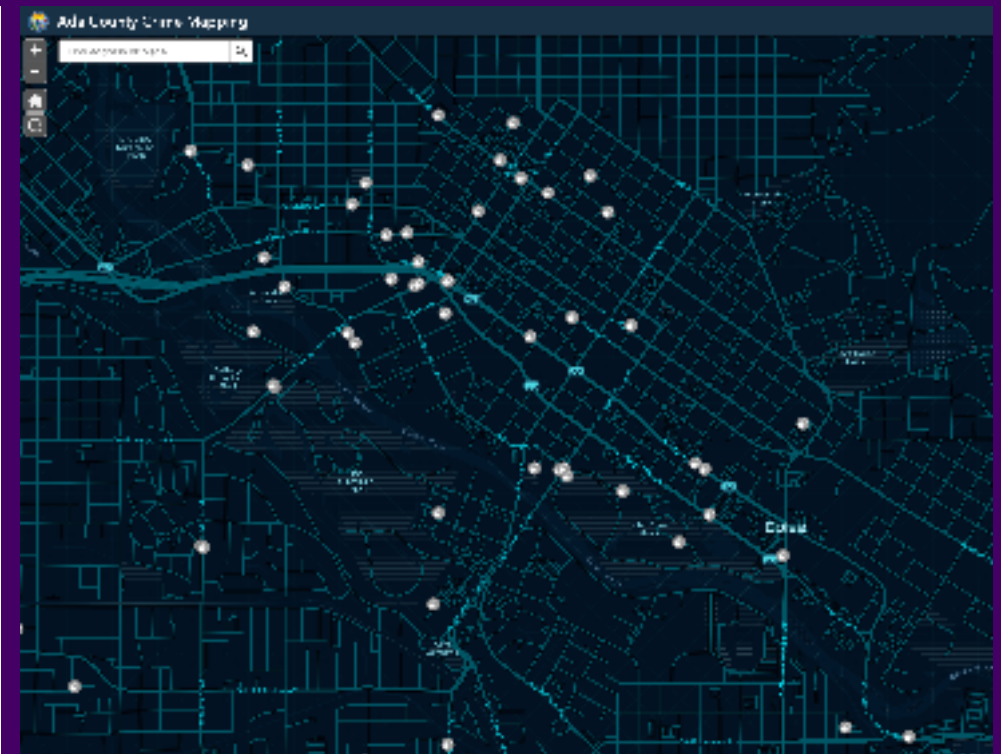
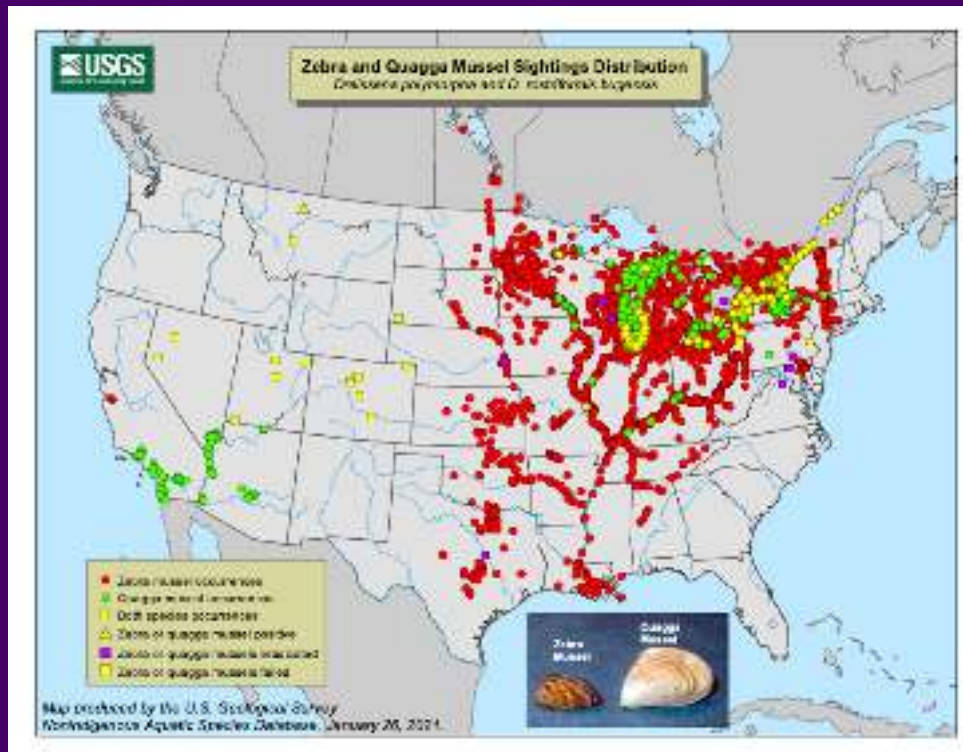
- Location
- Place
- Region
- Movement
- Human-
Environment
Interaction



WGBH Educational Foundation

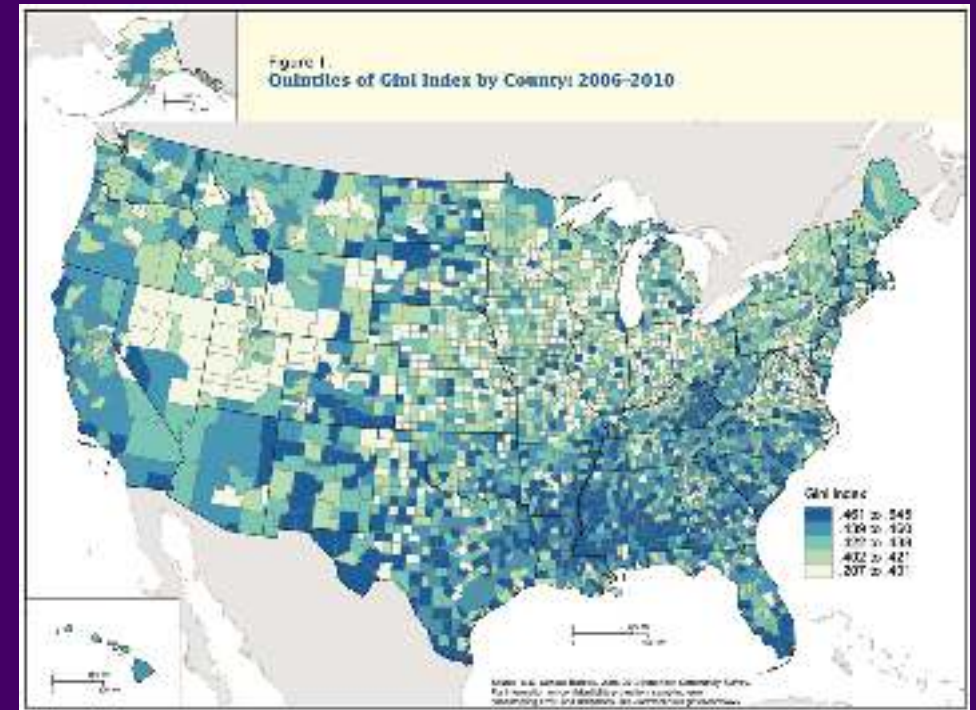
Location

The place (on Earth) of a particular geographic feature



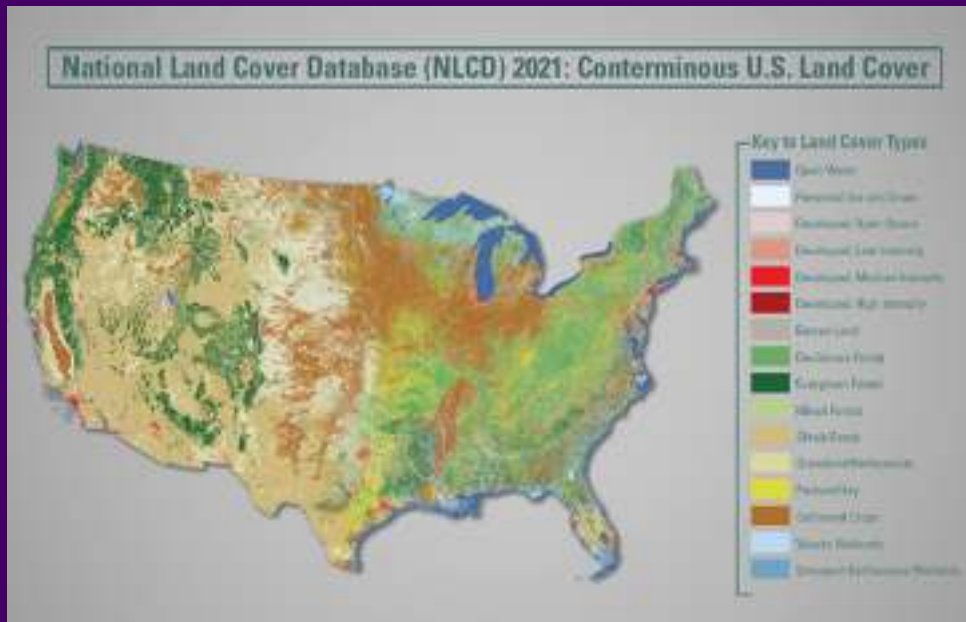
Place

What is a location *like*?



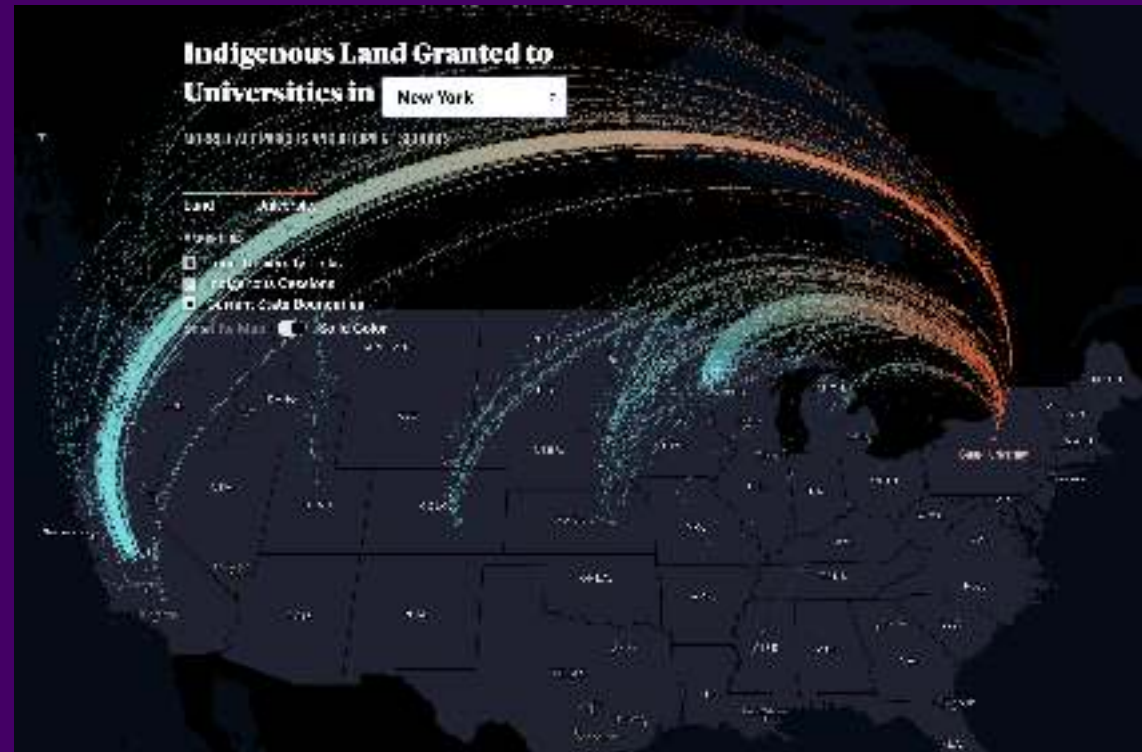
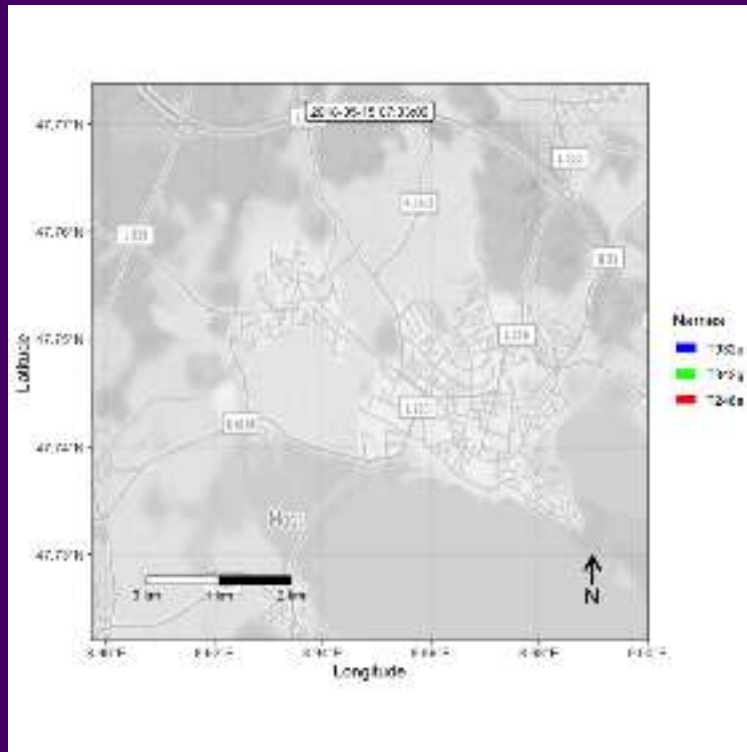
Region

What attributes to different geographies share? What distinguishes them?



Movement

How do genes, individuals, populations, ideas, goods, etc traverse the landscape.



Human-Environment Interactions

How do people relate to and change the physical world to meet their needs?

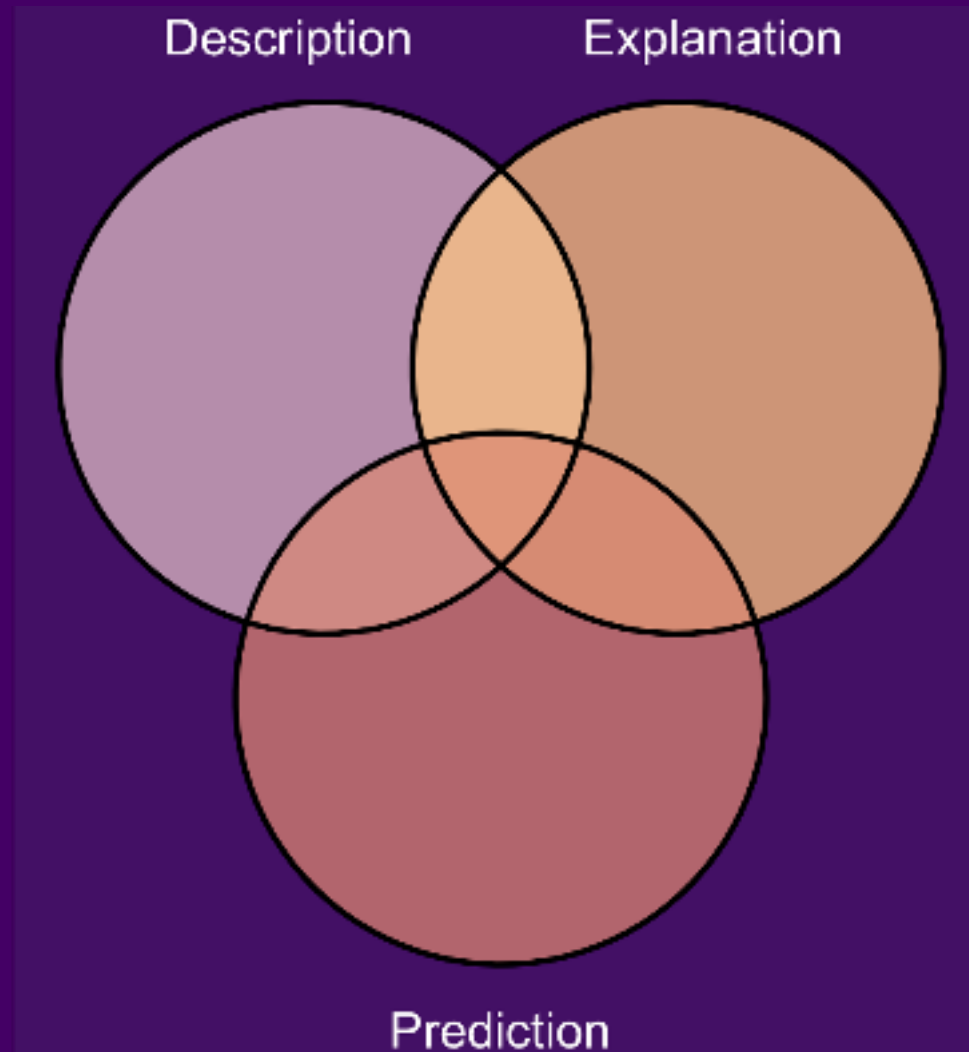


Towards *quantitative* spatial analysis

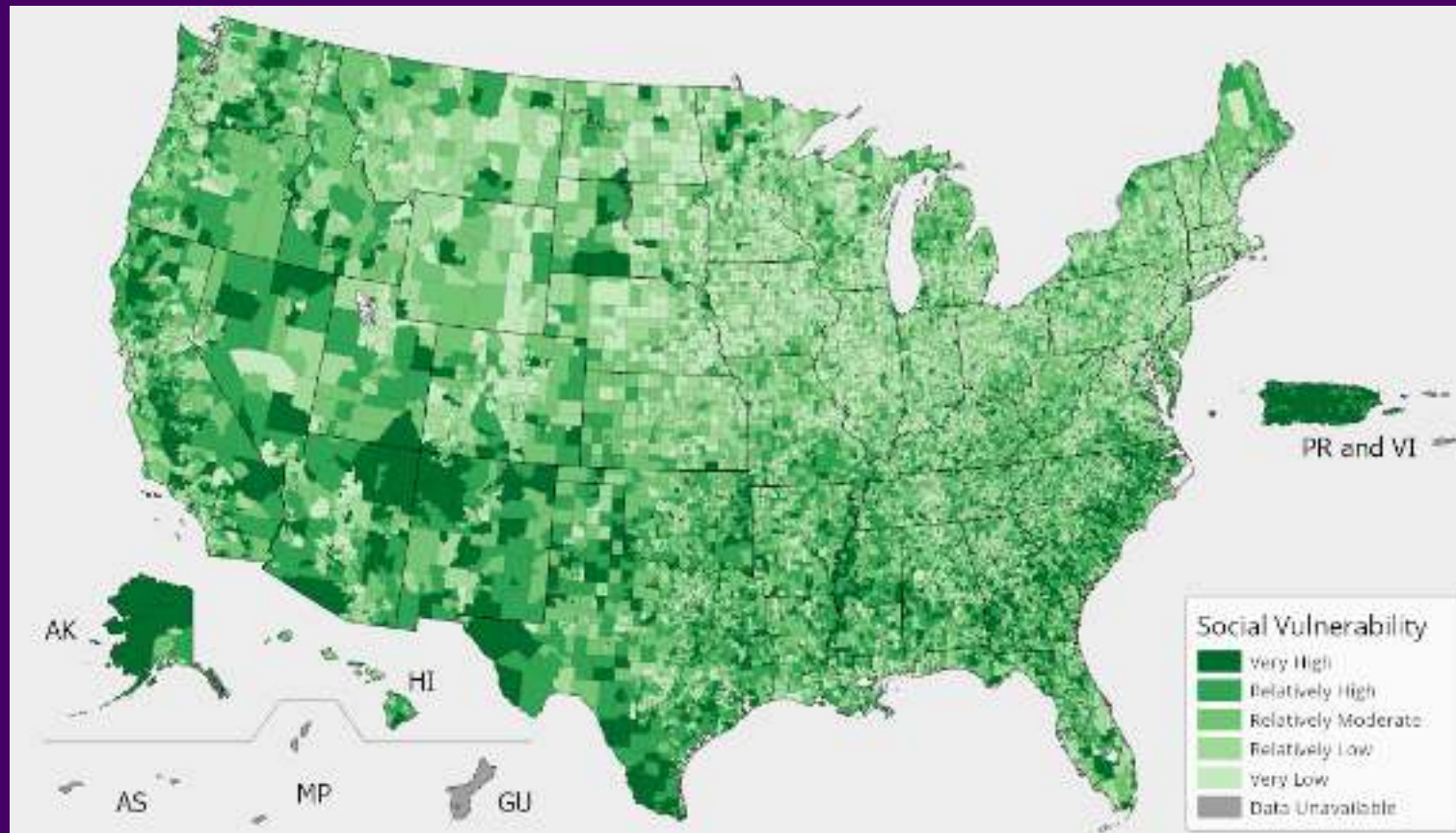
‘everything is usually related to all else but those which are near to each other are more related when compared to those that are further away’.

Waldo Tobler

3 “Faces” of Analysis



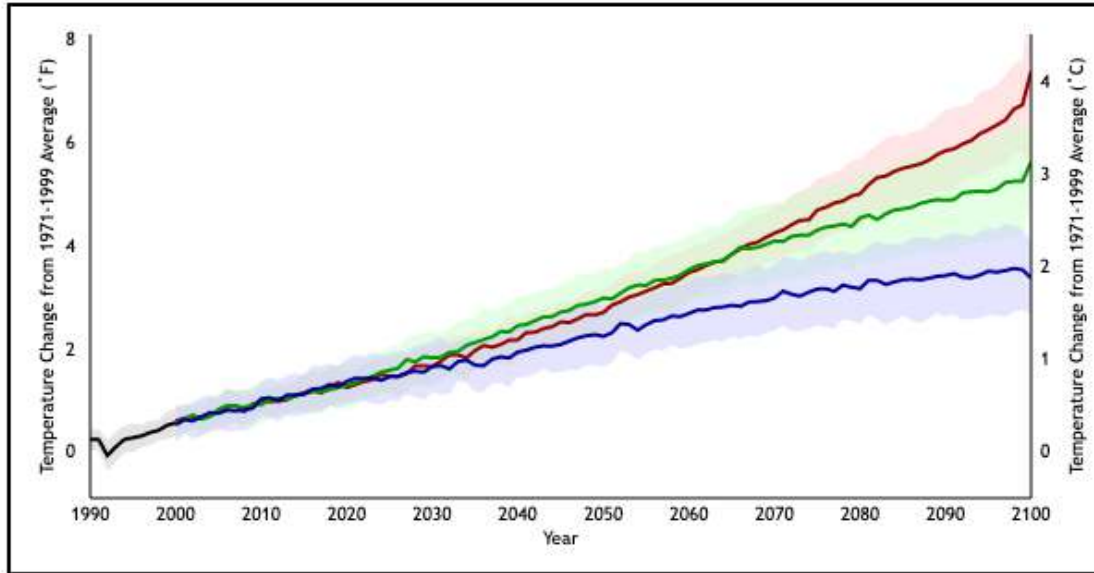
Description



Explanation and Inference

- **Cognitive Description:** collection ordering and classification of data
- **Cause and Effect:** design-based or model-based testing of the factors that give rise to geographic distributions
- **Systems Analysis:** describes the entire complex set of interactions that structure an activity

Prediction



- Extend description or explanation into unmeasured space
- Stationarity: the rules governing a process do not *drift* over space-time

Class Details

Logistics

- Meet on Mondays and Wednesdays
- ~40 min lecture, 30 min practice
- 5 major sections
- A note about readings

“Office Hours” immediately following class

Course Webpage

<https://isdrrfall25.classes.spaseslab.com/>

- Syllabus
- Schedule
- Lectures
- Assignments
- Resources

Assignments

Check out the syllabus for more on grading!

- **Self-reflections (2x)**
 - Your goals for the course
 - Evaluation criteria
- **In-class exercises (30x)**
 - Problem solving
 - Reproducible workflows
 - Muscle memory
- **Final project (1st draft, final draft):** Practice a full analysis workflow; Integrate analysis & visuals to tell a story
- **Homework Assignments (5x)**
 - Integrate skills from the section
 - Practice visualization
 - Build version control habits
- **Code Revisions (5x)**
 - Common issues
 - More extensive feedback

Wrapup

Checking in

1. What can I clarify about the course?

ATLAS ORBIS TERRARUM
MERIDIANO TABB,

DELINEATIO SINGULARI
RUDOLPHI

RATIONE ACCOMMOD
ASTRONOMICARUM



End

